19 A metal wire obeys Hooke's law and has a Young modulus of 2.0×10^{11} Pa. The wire has an original length of 1.6 m and a diameter of 0.48×10^{-3} m.

What is the spring constant of the wire?

A $7.2 \times 10^{3}\,\text{N m}^{-1}$

B $2.3 \times 10^{4}\,\text{N m}^{-1}$

C $2.9 \times 10^{4}\,\text{N m}^{-1}$

D $9.0 \times 10^{4}\,\text{N m}^{-1}$